



Inventra AI

MVP EXPERIENCE BLUEPRINT

The 30-Second Wow Moment

Designing the first AI Operating Copilot for SMEs

A product blueprint for the AIM Competition MVP – built to make a judge understand, in half a minute, that this is not another inventory system.

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What this blueprint covers

Eight parts, from the demo a judge sees to the operating system it becomes.

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PART 1

The Design Thesis — Win the Room in 30 Seconds

The judge starts the demo with a mental model: “another inventory tool.” Our entire job in the first 30 seconds is to break it.

Every POS vendor demo opens the same way: a product list, a stock count, a sales chart. If we open like that, we have already lost — we look like a cheaper version of something the judge has seen ten times. So we do the opposite. **We do not show data. We show a decision.**

The user drops a messy spreadsheet onto the screen, and a few seconds later Inventra AI hands back a one-page briefing written like a consultant’s memo: a plain-language verdict, the three things that need attention this week, and a ranked action plan — each action carrying a dollar figure. Then a chat box: *“Ask me anything about your business.”*

THE THREE THINGS A JUDGE MUST FEEL

1 · It read our real file.

Not a clean template — our actual export, with blank cells and mixed Khmer/English names.

2 · It has an opinion.

Not charts to interpret — a written verdict and a recommendation.

3 · It counted the money.

Every risk and opportunity is quantified: dollars at stake, dollars to gain.

WHAT WE DELIBERATELY DO NOT SHOW FIRST

Stock levels. Product tables. Reports. Reorder points. These are table stakes – necessary in the product, fatal as the opening move. They can appear later, once the judge has re-categorised us.

THE 30-SECOND TEST

If, after the demo, a judge cannot turn to the person next to them and say “**it read their data and told them what to do – and what it was worth**”, the demo failed. Every screen below is engineered to produce exactly that sentence.

PART 2

Hero Demo Scenario

One file. Thirty seconds. A deliberate, rehearsed sequence – and it also runs live on a judge’s own data.

The file. `shop_march.xlsx` – one sheet, 3,412 rows, 92 days of sales, product names in mixed Khmer and English, a scatter of blank cells, no tidy header formatting. Chosen to look exactly like what a real mini-mart owner would export and email to themselves.



Figure 2.1 – The 30-second sequence. Deterministic, rehearsed, and repeatable on any real export. Full detail in the table below.

Time	On screen	What the judge thinks
0:00	Presenter drags <code>shop_march.xlsx</code> onto a near-empty screen. One line of copy: "Drop your sales export."	"Okay, an import screen. Standard."
0:03	"Reading your business..." Counters animate: 3,412 rows · 287 products · 92 days · 6 categories . Column mapping appears — Product, Qty, Unit price, Date — each with a ✓ and a confidence score.	"It figured out my messy columns by itself. It didn't make me map anything."
0:08	The screen dissolves into The Business Brief . A headline verdict in plain language. Health score 72/100 (▲4) . Subline: "3 things need you this week."	"Wait — this is a written briefing. Not a dashboard."
0:12	Risks card animates in: ● Coca-Cola 1.25L stocks out in ~4 days (delivery takes 2). ● \$430 frozen in 6 products with no sale since January. ● 68% of revenue rides on one supplier.	"It's telling me what's about to go wrong — before it does."
0:16	Opportunities card: ● Iced coffee +34% month-on-month and under-stocked. ● Water + instant noodles bought together 61% of the time — not bundled. ● Fridays do 2.1× a weekday; you sell out by 3pm.	"It found money I didn't know was there."
0:20	Action Plan : a ranked list. Each row = action + projected impact + [Do this] . "Reorder Coke today — saves ~\$70." "Clear 6 dead SKUs — frees \$430 in ~10 days." "Bundle water + noodles — +\$30/wk." Footer: Projected upside this week: +\$180 revenue · \$430 cash freed.	"It did the thinking and put a number on every move."
0:25	Presenter types into the chat: " <i>What should I buy with \$500 this week?</i> " → 2 seconds → a costed list with reasoning: Coke \$180, noodles \$150, water \$90, iced coffee \$80 — snacks skipped (18 days of cover).	"I can just... ask it. Like a colleague."
0:30	Presenter: " We never opened an inventory screen. This isn't an inventory app — it's the analyst that reads one. " Then: "Hand us your file — we'll run it live."	The re-categorisation is complete.

THE LIVE VARIANT

The scripted run builds confidence; the **live run on a judge's own spreadsheet** wins it. Same pipeline, same 8 seconds. Offer it every time — the willingness to run unseen data is itself the proof.

PART 3

The AI Business Brief

The core artifact. A one-page briefing that reads like a sharp analyst wrote it overnight — not a dashboard, a *document*.

The Brief is what makes Inventra feel like a person rather than a panel of charts. It is generated on upload and refreshed every morning. It is scannable in 60 seconds and it always ends in three actions or fewer.

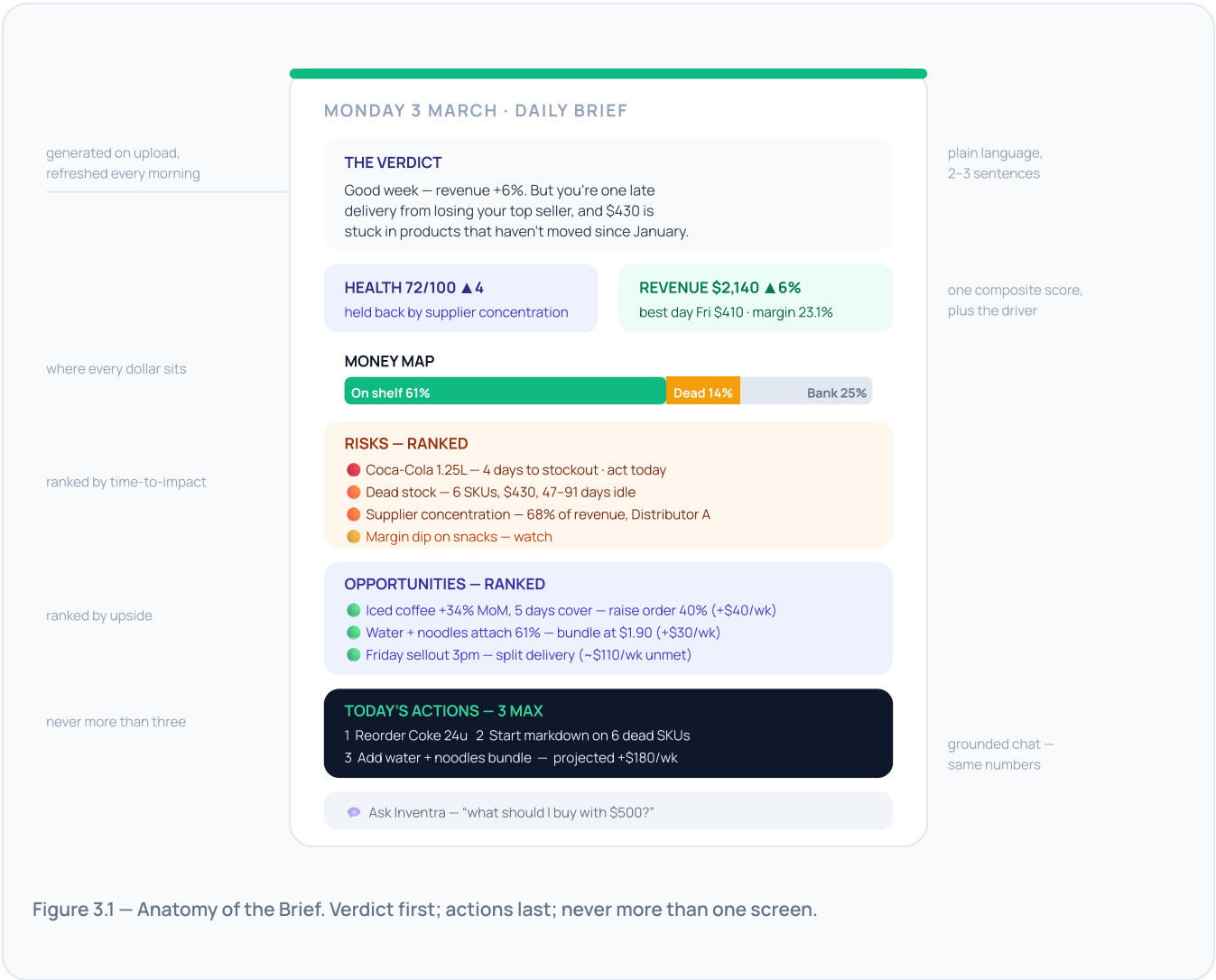


Figure 3.1 — Anatomy of the Brief. Verdict first; actions last; never more than one screen.

3.1 The eight blocks

Block	What it contains
The Verdict	2–3 sentences in plain language. The single most important paragraph — what’s going well, what’s at risk, what to do about it.
Health Score	A 0–100 composite of revenue trend, margin, inventory efficiency, cash position, and concentration — shown with its weekly delta and the one factor moving it most.
Business Health	Four tiles — Revenue, Margin, Inventory efficiency, Cash position — each with direction and a one-line “why.”
The Money Map	A single bar showing where cash is right now: healthy stock on the shelf, capital trapped in dead stock, cash available in the bank.
Risks	Ranked, five max, each with a severity (● today / ● this week / ● watch) and a time-to-impact.
Opportunities	Ranked, five max, each with an upside estimate in dollars per week.
Today’s Actions	Three max, one-tap, each carrying its projected impact. This is the part the owner acts on.
Ask Inventra	A chat box grounded in the same computed numbers — no new figures invented in conversation.

3.2 Worked example — Sophea’s Brief, Monday 3 March

THE VERDICT

“Good week — revenue up 6% to \$2,140, mostly from beverages. But you’re one late delivery from losing your top seller, and \$430 is sitting in products that haven’t moved since January. Fix those two and next week is your best month yet.”

Health	72 / 100 (▲4) — up on margin recovery; capped by supplier concentration
Revenue	\$2,140 this week (▲6% vs 4-week average) · best day Friday \$410 · slowest Tuesday \$190
Margin	23.1% (▲0.4 pt) — less snack discounting than last week
Inventory efficiency	14 days average cover (healthy band 10–21) · 6 SKUs above 45 days
Cash	\$1,180 available · \$430 trapped in dead stock · this week’s reorder plan needs \$610
Risks	● Coca-Cola 1.25L — 4 days to stockout, lead time 2 ● Dead stock \$430 / 6 SKUs ● 68% revenue from Distributor A
Opportunities	● Iced coffee +34% MoM, 5 days cover ● Water + noodles attach 61% ● Friday demand exceeds capacity, ~\$110/wk unmet
Today’s actions	Reorder Coke 24u · Start markdown on 6 dead SKUs · Launch water + noodles bundle — projected +\$180/wk revenue, \$430 cash freed

PART 4

AI Risk Detection

Detect the risks a busy owner cannot see — because they are spread across time, categories, and systems that never talk to each other.

An owner standing behind the till sees today. They do not see a category that has declined for three weeks straight, a supplier whose lead time has quietly crept from two days to five, or the fact that next week’s reorder plan is larger than the cash in the drawer. Inventra reads across all of it and ranks what matters.

INVENTORY RISK

Risk	Signal Inventra reads	Example alert
Imminent stockout on an A-item	Velocity × days-of-cover falls below lead time + safety buffer	"Coca-Cola 1.25L — 4 days left, delivery takes 2. Reorder today."
Silent stockout	On-hand at 0 for >72h while the SKU is still "active" and had steady sales	"Nescafé 3-in-1 has been out for 5 days — you're losing ~\$12/day and may not have noticed."
Expiry-bound overstock	Cover > shelf-life remaining, for perishables	"You have 60 days of yogurt cover and it expires in 21. Promote or return 40 units."
Dead stock	Zero sales for 45+ days with capital > threshold	"6 SKUs, \$430, no sale since January. Clear them."
Feast-or-famine SKU on a fixed reorder point	High demand variance managed by a static rule	"Ice cream swings 5× week to week — your fixed reorder causes both stockouts and spoilage."
Shrinkage signal	Units sold consistently < units depleted from stock	"Cigarettes: 18 units left the shelf, 14 were rung up, 4 weeks running. Investigate."

REVENUE RISK

Risk	Signal	Example alert
Revenue concentration	Top 5 SKUs or top 2 days > ~60% of revenue	"68% of your revenue is 5 products. If one supplier fails, most of your income does too."
Shrinking basket	Average items-per-transaction trending down 3+ weeks	"Baskets dropped from 3.1 to 2.6 items. Customers are buying less per visit — check placement and bundles."
Quietly declining category	Category revenue down each week for 3 weeks, masked by growth elsewhere	"Household goods have fallen 4 weeks running — hidden because beverages grew."
Underpriced fast-mover	Very high velocity + margin below category norm	"Iced coffee sells out daily at 23% margin vs 31% category. You're leaving ~\$25/wk on the table."
Recurring day-part sellout	Same SKUs hit 0 at the same time each day/week	"You run out of bread every Friday by 2pm — ~\$40 of demand walks out unserved."

SUPPLIER RISK

Risk	Signal	Example alert
Single-supplier dependency	One supplier > 50% of purchase value or of revenue-linked SKUs	"Distributor A supplies 68% of your revenue. Add a second source for beverages."
Creeping lead time	Order-to-delivery gap increasing order over order	"Distributor A now takes 5 days, not 2. Your safety stock is set for 2 — raise it or switch."
Price drift	One supplier's unit costs rising faster than others for comparable goods	"Your snack supplier's prices are up 9% in 2 months; the market is up 3%."
Falling fill rate	Delivered quantity < ordered quantity, trend	"Last 4 orders arrived 12% short. You're planning around stock you never receive."
MOQ-forced overstock	Minimum order quantity > sensible cover for the SKU	"The 48-unit case gives you 70 days of cover on a slow item — split the order or drop the SKU."

CASH FLOW RISK

Risk	Signal	Example alert
Rising trapped-capital ratio	Share of stock value in slow/non-moving SKUs increasing	"22% of your stock value is now slow-moving, up from 14%. Cash is freezing on the shelves."
Reorder exceeds cash	Recommended purchase plan > cash on hand	"This week's ideal reorder is \$780; you have \$610. Here's the \$610 version that protects your top sellers."
Predicted shortfall on a date	Projected outflows > inflows on a known day	"You'll be about \$180 short on the 15th when the supplier invoice is due. Delay the snack reorder to bridge it."
Margin compression trend	Blended margin declining while revenue flat/up	"You're selling more but keeping less — margin down 1.4 pts over a month, mostly discounting."
Seasonal cash crunch ahead	Historical pattern of a demand spike needing stock 3 weeks out	"Khmer New Year needs ~\$900 of extra stock in 3 weeks. Start setting aside \$300/wk now."

PART 5

AI Opportunity Detection

The same data that shows what's wrong also shows what's possible. Inventra ranks the upside by dollars-per-week and effort.

Opportunity	Pattern Inventra reads	Example & estimated upside
Accelerating product	Velocity rising and the rate of rise is increasing, while stock cover is thin	"Iced coffee +34% MoM, 5 days cover — raise the order 40%." +\$40/wk
Emerging category	New or minor SKUs steadily taking basket share	"Ready-meals went from 2% to 7% of baskets in 6 weeks — widen the range." +\$60/wk
Early seasonal ramp	Current demand curve leading the same period in prior years	"Cold drinks are ramping 10 days earlier than last year — pre-stock before the heat." avoids ~\$120 lost sales
Frequently bought together	Two SKUs co-occur in a high share of transactions, not merchandised together	"Water + instant noodles: 61% attach. Bundle at \$1.90 or shelve them together." +\$30/wk
Trade-up gap	Most buyers take the small size; the large earns more margin per sale	"82% buy the 400ml water. Push the 1.5L at the counter — +\$0.15 margin each." +\$18/wk
Price headroom	High velocity, no volume drop across past price moves	"Iced coffee held volume through the last price rise — test +\$0.10." +\$22/wk
Range gap	Repeated stockouts + substitutions + "do you have X?" patterns point to a missing SKU	"You lose ~6 sales/week to not stocking large eggs — add the SKU." +\$25/wk
Day-part / day-of-week expansion	Demand consistently exceeds capacity in a specific window	"Fridays sell out by 3pm — split the delivery Wed + Fri." +\$110/wk
Dead-stock-to-cash	Trapped capital that could be redeployed into faster movers	"Clear the \$430 of dead stock, put it into your top 3 SKUs turning 3x faster." +\$35/wk on redeployed cash
Supplier consolidation	Volume split across suppliers that one could serve at a better rate	"Combine beverage orders with one distributor for a ~4% volume discount." +\$28/wk
Loyalty pattern	Identifiable repeat buyers concentrated on a few SKUs	"~20 regulars buy coffee daily — a simple '10th free' card locks the habit." retention
Event / calendar demand	Paydays, holidays, local events, weather — historically move specific SKUs	"Payday is the 25th — premium items sell 1.6x. Stock up on the 23rd." +\$70 that week

PART 6

AI Action Plan

Reporting is not the product. Every insight resolves into a specific, costed action the owner can take this week.

A risk without an action is an anxiety. An opportunity without an action is a regret. Inventra never leaves an insight hanging — each one carries a recommended move, the reasoning, a projected dollar impact, and an effort tag.

Situation	Recommended action	Rationale & projected impact
Snacks over-stocked, beverages turning fast	"Reduce snack purchases by 20% and move that ~\$60/wk into beverages."	Beverages turn 3× faster; snacks carry 18 days of cover. +\$45/wk margin, shelf freed.
Top seller 4 days from stockout	"Reorder Coca-Cola 1.25L today — 24 units — not Thursday."	Delivery takes 2 days; stockout in 4. Avoids ~\$70 of lost sales.
\$430 in six products, no sale since January	"Mark those 6 SKUs down 30% this week; stop restocking them."	47–91 days idle. Recovers ~\$300 cash in ~10 days; permanent shelf back.
Iced coffee: high velocity, low elasticity	"Raise iced-coffee price by \$0.10."	Volume held through the last two price moves. +\$22/wk margin.
Water + noodles: 61% attach, not merchandised	"Create a water + instant-noodle bundle at \$1.90; shelve them together."	Already bought together most of the time. +8% attach, ~+\$30/wk.
Friday demand exceeds one delivery	"Split the weekly delivery into Wednesday + Friday."	You sell out Friday afternoons. Captures ~\$110/wk of unmet demand.
68% of revenue from one supplier	"Onboard a second beverage supplier for your top 5 SKUs."	Removes a single point of failure. ~5% price leverage + resilience.
Product X: 90 days, zero sales, \$70 tied up	"Stop restocking Product X. Clear the last units at cost."	No demand signal for a full quarter. \$70 cash + shelf recovered permanently.
Khmer New Year in 3 weeks	"Pre-stock the 12 festival SKUs now; set aside \$300/wk for 3 weeks."	Demand historically 2.2× for 10 days. Avoids stockout, captures the peak (~\$400).
Top 5 SKUs sit on a low shelf	"Move your five fastest movers to eye level."	Velocity vs shelf-position mismatch. +3–5% on those lines.
Reorder plan (\$780) exceeds cash (\$610)	"Use this \$610 plan — it fully covers your top 8 sellers and defers 3 slow SKUs."	Protects revenue while staying solvent. Zero stockout risk on A-items.
Nescafé silently out for 5 days	"Add Nescafé 3-in-1 to today's order; set a low-stock alert at 6 units."	Steady seller, nobody noticed. Stops ~\$12/day bleed.



Figure 6.1 – The action lifecycle. The weekly feedback loop – showing what the actions were worth – is the retention engine.

PART 7

Differentiation — “I Already Have a POS. Why Inventra?”

Because a POS is a system of record. Inventra is a system of decisions. They are not the same category.

Your POS is your business’s memory. Inventra AI is its judgment. One records what happened. The other reads across all of it and tells you what to do next — and what it’s worth.

Inventra does not replace the POS. It sits on top of it — and on top of Excel, supplier messages, and anything else the business already produces. Keep the POS. Add a brain.

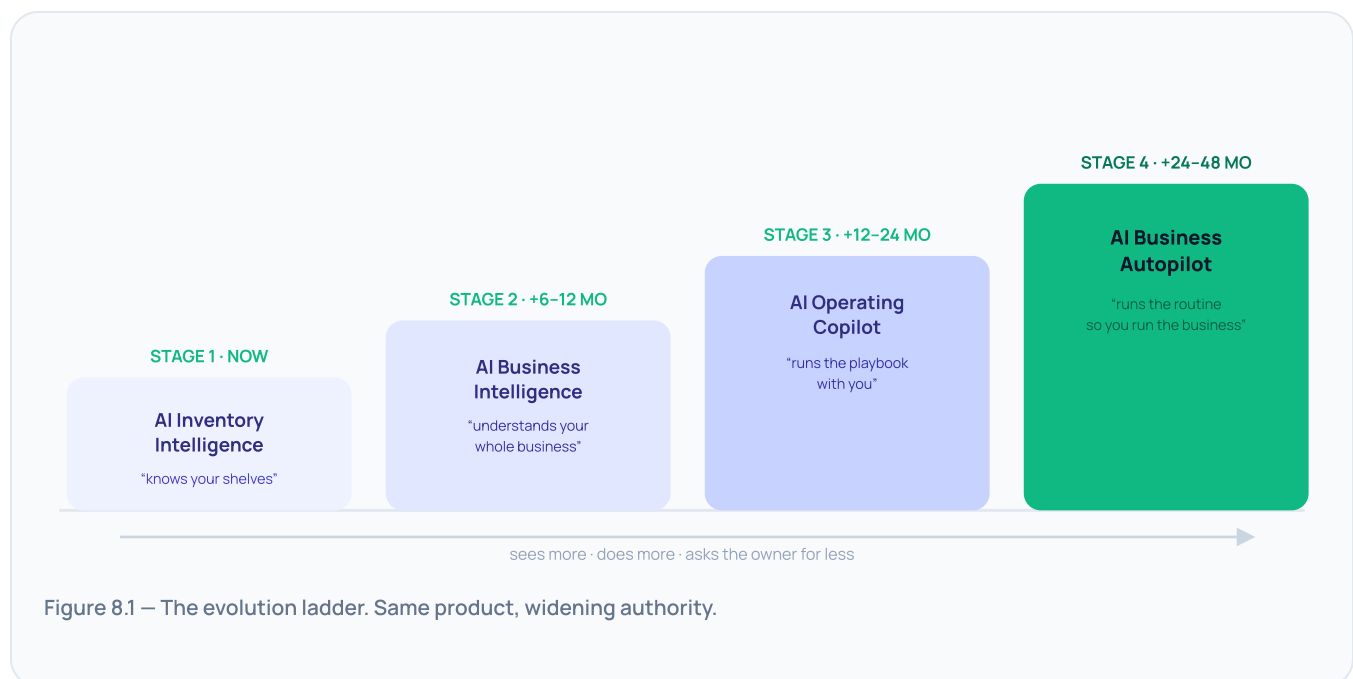
Dimension	POS / inventory system	Inventra AI
Core job	Record transactions and stock	Read across all the data and decide what to do
Question it answers	“What did I sell?”	“What should I do next — and why?”
Output	Reports and screens you interpret	A briefing written for you, plus a ranked action plan
Time orientation	Backward — what already happened	Forward — predictive and prescriptive
Alerts	Reactive thresholds (“low stock”)	Risk detected early, ranked by money and time-to-impact
Scope of view	One store’s transactions	Sales + margin + cash + suppliers + seasonality, together
Who does the analysis	You do — the POS needs an analyst to be useful	Inventra is the analyst
Interface	Menus and tables	A one-page brief and a chat box

1. **The POS never tells them what to do.** It shows the numbers and stops. The owner still has to be the analyst at 10pm.
2. **The POS only sees transactions.** It cannot connect a supplier's slipping lead time to next month's cash gap to the category that's quietly declining. Inventra holds all of that in one view.
3. **The POS makes them work for the insight.** Inventra delivers the insight – and the action, and the dollar figure – before they've finished their coffee.

PART 8

Future Vision — From Inventory Intelligence to Business Autopilot

Four stages. At each one Inventra sees more, does more, and asks the owner for less – but the owner always sets the goals.



Stage	Question it answers	What it can see	What it does	The owner's role
1 – Inventory Intelligence (MVP)	"What's happening on my shelves?"	Sales history and stock levels	Forecasts demand, flags dead stock and stockouts, suggests reorders	Makes every buying decision
2 – Business Intelligence	"How is my business really doing?"	+ margin, cash position, suppliers, seasonality	Writes the daily Brief; detects risk and opportunity across the whole business; costs every action	Decides, using the Brief
3 – Operating Copilot	"What should I do – and can you help me do it?"	+ pricing, promotions, supplier catalogues, staffing	Conversational and proactive; drafts purchase orders, price changes and promotions for one-tap approval	Approves; sets strategy
4 – Business Autopilot	"Run the routine; flag the exceptions."	+ bank feed, e-commerce, multi-branch	Executes routine decisions inside owner-set budgets and guardrails – auto-reorder, dynamic pricing, supplier switching – and surfaces only exceptions	Sets goals and guardrails; handles exceptions

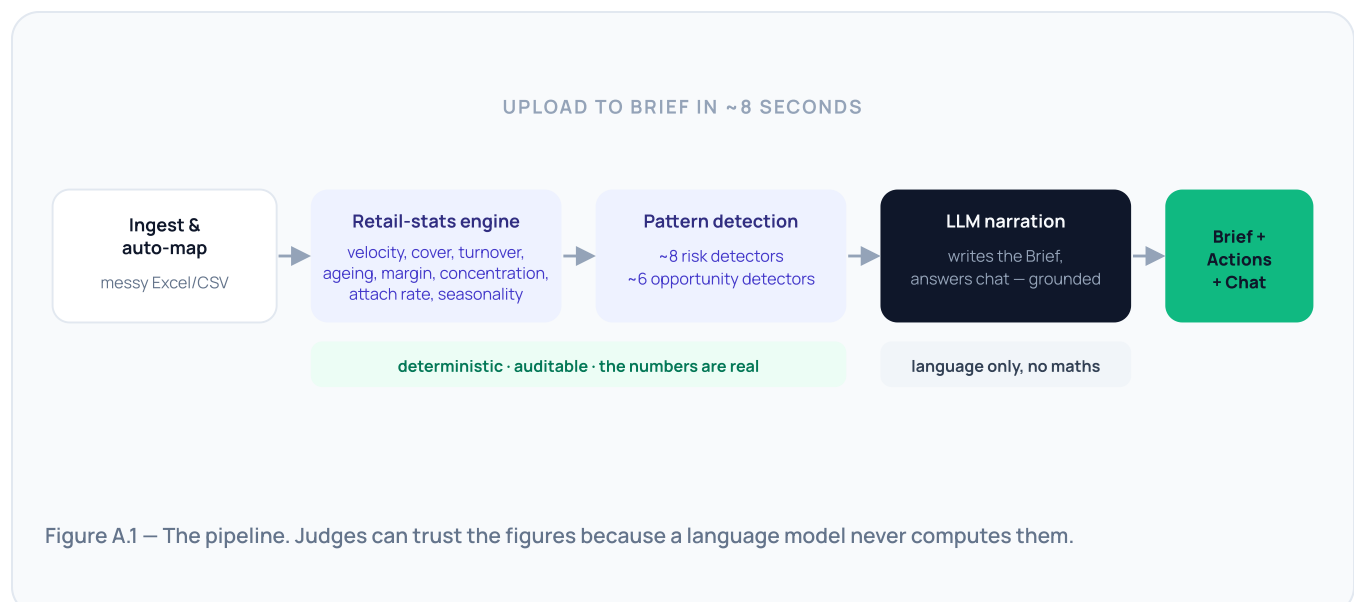
THE THROUGH-LINE

We are not building software that manages inventory. We are building the operating system that runs the business – starting with the one decision an SME owner makes every single week: **what to buy**.

APPENDIX

How the Wow Actually Works

The magic is a pipeline. Deterministic maths does the analysis; the language model writes the story – it never invents a number.



IN SCOPE FOR BUILD WEEK

File ingest + auto-mapping · retail-stats engine · ~8 risk detectors · ~6 opportunity detectors · Brief generator · grounded chat · action list with projected impact · the 30-second demo flow.

DELIBERATELY OUT OF SCOPE (FOR NOW)

Live POS integration · write-back (auto-drafting real POs) · multi-branch consolidation · bank-feed cash forecasting · autonomous execution. These are Stages 3–4, not the MVP.

Do not build another inventory system. Build the analyst that reads one — and, eventually, the operating system that runs the shop.



Inventra AI

MVP Experience Blueprint · v1.0 · for the AIM Competition